

# PAPER\_284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product ( $\Phi_{dm}$ )

## SFR Growth × Photoevaporation Erosion Multiplicative Coupling

**Classification:** UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics

**System:** M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region

**Session:** 80 | **Module:** M16\_UQFF\_MODULE.cpp (22nd C++ UQFF module)

**Author:** Daniel T. Murphy | **Date:** March 2026

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### Abstract

This paper introduces the **Dual Mass Co-Action Product** ( $\Phi_{dm}$ ) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a **multiplicative** product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate  $SFR = 1 \text{ M}\odot/\text{yr}$  over initial gas mass  $M_0 = 1200 \text{ M}\odot$ , and maximum photoevaporation fraction  $E_0 = 0.3$  (30%), the multiplicative form produces a 24.3% reduction in  $\Phi_{dm}$  relative to the additive approximation at  $t = 5 \text{ Myr}$ . This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

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## 2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

1. **Star Formation Accretion** — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by  $SFR\_rate \times t$ .
2. **Photoevaporation Erosion** — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction  $E_{rad}(t)$ .

The **additive approximation** (used in prior UQFF modules):

$$\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}$$

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

$$\Phi_{dm}(t) = (1 + SFR\_rate \times t) \times (1 - E_{rad}(t))$$

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16’s “Pillars of Creation”).

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## 3. Mathematical Formulation

### 3.1 Parameters (M16 Eagle Nebula)

| Parameter             | Value                                       | Description                                             |
|-----------------------|---------------------------------------------|---------------------------------------------------------|
| $M_0$                 | $2.387 \times 10^{33}$ kg (1200 $M_\odot$ ) | Initial nebula gas mass                                 |
| SFR                   | 1 $M_\odot$ /yr                             | Star formation rate                                     |
| SFR_rate              | $2.639 \times 10^{-11}$ s $^{-1}$           | = SFR / ( $M_0/M_\odot$ ) / ( $3.156 \times 10^7$ s/yr) |
| $\tau_{\text{erode}}$ | $9.468 \times 10^{13}$ s (3 Myr)            | Photoevaporation e-folding time                         |
| $E_0$                 | 0.3                                         | Maximum photoevaporation fraction                       |
| $g_{\text{base}}$     | $1.454 \times 10^{-12}$ m/s $^2$            | $G \times M / r^2$ at $r = 3.31 \times 10^{17}$ m       |

### 3.2 Dual Co-Action Product

$$M_{sf}(t) = \text{SFR\_rate} \times t$$

$$E_{rad}(t) = E_0 (1 - e^{-t/\tau})$$

$$\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})$$

### 3.3 Dynamic Gravity Term

$$g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)$$

### 3.4 Multiplicative-Additive Gap

The gap relative to the additive form:

$$\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})$$

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

## 4. Numerical Results at t = 5 Myr

$$t = 5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$$

| Quantity                     | Value                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------|
| M_sf_frac                    | SFR_rate $\times$ t = $2.639 \times 10^{-11} \times 1.578 \times 10^{14} = \mathbf{4164.8}$ |
| E_rad                        | $E_0 \times (1 - \exp(-5/3)) = 0.3 \times 0.8110 = \mathbf{0.2433}$                         |
| $\Phi_{dm}$ (multiplicative) | $(1 + 4164.8) \times (1 - 0.2433) = 4165.8 \times 0.7567 = \mathbf{3151.9}$                 |
| $\Phi_{dm}$ (additive)       | $(1 + 4164.8) - 0.2433 = \mathbf{4165.6}$                                                   |
| <b>gap_mult_add</b>          | $-(4164.8 \times 0.2433) = \mathbf{-1013.3}$<br>(24.3% less)                                |
| $g_{dyn}(5 \text{ Myr})$     | $1.454 \times 10^{-12} \times 3151.9 = \mathbf{4.583 \times 10^{-9} \text{ m/s}^2}$         |

The multiplicative gap of  $-1013.3$  confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

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## 5. Connection to UQFF 2.0 $g_{\text{total}}$

In the full M16 UQFF 2.0 equation:

$$g_{\text{total}}(r, t) = [g_{\text{dyn}}(t) + U_{g,\text{sum}}(26) + \Lambda + Q + L + F + g_{\text{exp}}] \times \text{corr}_{SC}$$

The  $\Phi_{\text{dm}}$  product modulates only the dynamic base gravity term  $g_{\text{dyn}}$ . The 26-layer Triadic ( $U_{g,\text{sum}}$ ), cosmological  $\Lambda$ , quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of  $\Phi_{\text{dm}}$  — the modulation is cleanly scoped to the time-evolving mass component.

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## 6. UQFF Historical Distinction

| Module                                       | SFR Term                                     | Erosion Term                  | Form                  |
|----------------------------------------------|----------------------------------------------|-------------------------------|-----------------------|
| Session 55 CP3<br>M16EagleNebulaRadiationSFR | $g_{\text{base}} \times (1 + M_{\text{sf}})$ | $-E_{\text{rad}}$             | <b>Additive</b>       |
| <b>This paper<br/>(PAPER_284)</b>            | $(1 + M_{\text{sf}})$                        | $\times (1 - E_{\text{rad}})$ | <b>Multiplicative</b> |

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

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## 7. Wolfram KB Term

M16UQFF:Phi\_dm=(1+SFR\_rate\*t)\*(1-E\_0\*(1-Exp[-t/tau])); SFR\_rate=2.639e-11/s; M\_sf\_frac=SFR\_rate\*t [PAPER\_284]

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## 8. Cross-References

- **PAPER\_285:** Erosion Saturation Half-Time ( $t_{\text{half}}$ ,  $\Delta g_{\text{Max}}$ )
- **PAPER\_286:** Nebular Friedmann Redshift ( $\kappa_{\text{neb}}$ ,  $z=0.0015$ )
- **M16\_UQFF\_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
- **CondensedPhysics3.py:** M16DualMassCoActionProductCalculator

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                        | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|------------------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                       | 1/137.036                              | PDG 2024                            | ✓ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)                | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | ✓ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | ✓ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                               | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*